Theory

Functions are reusable blocks of code that perform a specific task. In R, user-defined functions are created using the function() keyword and can accept parameters, perform computations, and return results.

R Function Syntax:

- function_name <- function(arg1, arg2, ...) { body; return(value) }
- Arguments can have default values: function(x, n = 2) { ... }
- return() sends one or more values back to the caller; use list() to return multiple values.
- Functions can call built-in R functions like mean(), sd(), var(), round() inside their body.

Key Concepts:

- Central Tendency — Mean: average value; Median: middle value; Mode: most frequent value.
- Spread / Dispersion — Variance: average squared deviation; Std Dev: sqrt(variance); Range: max-min; IQR: Q3-Q1.
- Area of Circle — Formula: A = pi * r^2. R provides the constant 'pi' built-in (3.141593).

Problem 5a — Measures of Central Tendency and Spread

A user-defined function stats_summary(x) is created that accepts a numeric vector and computes all measures of central tendency (Mean, Median, Mode) and spread (Range, Variance, Std Dev, IQR, Min, Max, Length). It prints a formatted report and returns a named list.

Expected Values for x = c(2, 4, 10, 11, 5, 7)

Measure	Formula / Function	Result
Mean	sum(x) / length(x)	6.5
Median	middle value (sorted)	6.0
Mode	most frequent element	2
Min	min(x)	2
Max	max(x)	11
Range	range(x)	2 to 11
Variance	var(x)	12.3
Std Deviation	sd(x)	3.5071
IQR	IQR(x) = Q3 - Q1	5.25
Length	length(x)	6

Code for Problem 5a — Function Definition and Function Call:

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5a_central_tendency.R

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```r
# Practical 5a: Function for Measures of Central Tendency and Spread
#              x = [2, 4, 10, 11, 5, 7]

# ── Define the function ──────────────────────────────────────
stats_summary <- function(x) {

  # Measures of Central Tendency
  mean_val <- mean(x)
  median_val <- median(x)

  # Mode (most frequently occurring value)
  freq_table <- table(x)
  mode_val <- as.numeric(names(which.max(freq_table)))

  # Measures of Spread
  range_val <- range(x)
  var_val <- var(x)
  sd_val <- sd(x)
  iqr_val <- IQR(x)
  min_val <- min(x)
  max_val <- max(x)
  length_val <- length(x)

  # Print Results
  cat("====================================\n")
  cat("   STATISTICAL SUMMARY REPORT\n")
  cat("====================================\n")
  cat("Input Vector   :", x, "\n")
  cat("------------------------------------\n")
  cat("CENTRAL TENDENCY\n")
  cat("  Mean         :", mean_val,   "\n")
  cat("  Median       :", median_val, "\n")
  cat("  Mode         :", mode_val,   "\n")
  cat("------------------------------------\n")
  cat("SPREAD / DISPERSION\n")
  cat("  Min          :", min_val,    "\n")
  cat("  Max          :", max_val,    "\n")
  cat("  Range        :", range_val,  "\n")
  cat("  Variance     :", round(var_val, 4), "\n")
  cat("  Std Dev      :", round(sd_val,  4), "\n")
  cat("  IQR          :", iqr_val,    "\n")
  cat("  Length       :", length_val, "\n")
  cat("====================================\n")

  # Return list of all computed values
  return(list(mean = mean_val, median = median_val, mode = mode_val,
              variance = var_val, sd = sd_val, range = range_val, IQR = iqr_val))
}

# ── Call the function ──────────────────────────────────────
x <- c(2, 4, 10, 11, 5, 7)
cat("Calling stats_summary() with x = c(2, 4, 10, 11, 5, 7)\n\n")
result <- stats_summary(x)
```

Output for Problem 5a — Statistical Summary Report:

Console ~/

```
> x <- c(2, 4, 10, 11, 5, 7)
> cat("Calling stats_summary() with x = c(2, 4, 10, 11, 5, 7)\n\n")
Calling stats_summary() with x = c(2, 4, 10, 11, 5, 7)

> result <- stats_summary(x)
====================================
   STATISTICAL SUMMARY REPORT
====================================
Input Vector   : 2 4 10 11 5 7
------------------------------------
CENTRAL TENDENCY
  Mean         : 6.5
  Median       : 6
  Mode         : 2
------------------------------------
SPREAD / DISPERSION
  Min          : 2
  Max          : 11
  Range        : 2  11
  Variance     : 12.3
  Std Dev      : 3.5071
  IQR          : 5.25
  Length       : 6
====================================
```

Code for Problem 5a — Verify Returned List and Cross-Check with summary():

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5a_verify.R

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```r
# Verify the returned list from stats_summary()

cat("\n--- Accessing Returned List Values ---\n")
cat("result$mean     :", result$mean,     "\n")
cat("result$median   :", result$median,   "\n")
cat("result$mode     :", result$mode,     "\n")
cat("result$variance:", round(result$variance, 4), "\n")
cat("result$sd       :", round(result$sd,       4), "\n")
cat("result$range    :", result$range,    "\n")
cat("result$IQR      :", result$IQR,      "\n")

# Visualise using boxplot and summary()
cat("\n--- Built-in summary() for cross-check ---\n")
print(summary(x))
```

Output for Problem 5a — Returned List Values and Built-in summary():

Console ~/

```
> cat("\n--- Accessing Returned List Values ---\n")

--- Accessing Returned List Values ---
result$mean     : 6.5
result$median   : 6
result$mode     : 2
result$variance: 12.3
result$sd       : 3.5071
result$range    : 2  11
result$IQR      : 5.25

> cat("\n--- Built-in summary() for cross-check ---\n")

--- Built-in summary() for cross-check ---
> print(summary(x))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
  2.000   4.250   6.000   6.500   9.250  11.000 
```

Problem 5b — Area of Circle Function

A user-defined function area_of_circle(radius) is created that accepts radius as a parameter, validates that it is positive, and computes the Area, Diameter, and Circumference of the circle using the built-in constant pi. The function is tested with radii = 5, 7.5, 1, and an invalid -3.

Formulae Used

Property	Formula	For r = 5
Diameter	$D = 2 \times r$	10 units
Area	$A = \pi \times r^2$	78.5398 sq. units
Circumference	$C = 2 \times \pi \times r$	31.4159 units

Code for Problem 5b — Area of Circle Function Definition and Test Cases:

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5b_area_circle.R

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```r
# Practical 5b: Function to Perform Area of Circle

# ── Define the function ──────────────────────────────────────
area_of_circle <- function(radius) {

  # Validate: radius must be positive
  if (radius <= 0) {
    cat("Error: Radius must be a positive number.\n")
    return(NULL)
  }

  # Compute area using formula: A = pi * r^2
  area <- pi * radius ^ 2

  # Compute circumference: C = 2 * pi * r
  circumference <- 2 * pi * radius

  # Compute diameter: D = 2 * r
  diameter <- 2 * radius

  # Print detailed output
  cat("====================================\n")
  cat("       CIRCLE PROPERTIES\n")
  cat("====================================\n")
  cat("  Radius          :", radius,"units\n")
  cat("  Diameter        :", diameter,"units\n")
  cat("  Area            :", round(area,4),"sq. units\n")
  cat("  Circumference   :", round(circumference,4),"units\n")
  cat("  Value of pi     :", pi, "\n")
  cat("====================================\n")

  return(list(radius = radius, diameter = diameter,
              area = area, circumference = circumference))
}

# ── Call the function with different radii ──────────────────
cat("--- Test 1: radius = 5 ---\n")
r1 <- area_of_circle(5)

cat("\n--- Test 2: radius = 7.5 ---\n")
r2 <- area_of_circle(7.5)

cat("\n--- Test 3: radius = 1 (unit circle) ---\n")
r3 <- area_of_circle(1)

cat("\n--- Test 4: Invalid radius = -3 ---\n")
area_of_circle(-3)
```

Output for Problem 5b — Circle Properties for All Test Cases:

```
Console ~/
> cat("--- Test 1: radius = 5 ---\n")
--- Test 1: radius = 5 ---
> r1 <- area_of_circle(5)
====================================
       CIRCLE PROPERTIES
====================================
  Radius         : 5 units
  Diameter       : 10 units
  Area           : 78.5398 sq. units
  Circumference  : 31.4159 units
  Value of pi    : 3.141593
====================================

> cat("\n--- Test 2: radius = 7.5 ---\n")

--- Test 2: radius = 7.5 ---
> r2 <- area_of_circle(7.5)
====================================
       CIRCLE PROPERTIES
====================================
  Radius         : 7.5 units
  Diameter       : 15 units
  Area           : 176.7146 sq. units
  Circumference  : 47.1239 units
  Value of pi    : 3.141593
====================================

> cat("\n--- Test 3: radius = 1 (unit circle) ---\n")

--- Test 3: radius = 1 (unit circle) ---
> r3 <- area_of_circle(1)
====================================
       CIRCLE PROPERTIES
====================================
  Radius         : 1 units
  Diameter       : 2 units
  Area           : 3.1416 sq. units
  Circumference  : 6.2832 units
  Value of pi    : 3.141593
====================================

> cat("\n--- Test 4: Invalid radius = -3 ---\n")

--- Test 4: Invalid radius = -3 ---
> area_of_circle(-3)
Error: Radius must be a positive number.
```

Conclusion

Thus, the following R programs on user-defined functions for Practical 5 were successfully written, executed in RStudio, and outputs were verified:

- Problem 5a: A function stats_summary(x) was defined to compute all measures of central tendency and spread for the vector x = c(2, 4, 10, 11, 5, 7). The function correctly calculated Mean = 6.5, Median = 6, Mode = 2, Variance = 12.3, Std Dev = 3.5071, IQR = 5.25, Range = 2 to 11, and Length = 6. The function returns a named list, allowing individual values to be accessed via result$mean, result$sd, etc. Results were cross-verified using R's built-in summary() function.
- Problem 5b: A function area_of_circle(radius) was defined using the formula A = pi * r^2. It includes input validation (returns error for negative radius), and also computes Diameter (D = 2r) and Circumference (C = 2*pi*r). The function was tested with four cases: r=5 (Area=78.5398), r=7.5 (Area=176.7146), r=1 (Area=3.1416), and r=-3 (error message printed correctly).

Both programs demonstrated the concept of user-defined functions in R — encapsulating logic into reusable, parameterized blocks that accept inputs, perform computation, print formatted output, and return structured results.